Question		Answer	Marks	AO	Guidance
11	(a)	because Sam used a larger sample **oe**	**B1** **[1]**	**2.4**	allow eg because he/she/they used more data eg the sample is bigger
11	(b)	0.76 **or** $\frac{38}{50}$ **oe** BC	**B1** **[1]**	**1.1**	mark the final answer
11	(c)	0.19	**B1** **[1]**	**3.3**	**FT** $\frac{their\ 0.76}{4}$

Question		Answer	Marks	AO	Guidance
11	(d)	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>21.52</td><td>20.19</td><td>7.11</td><td>1.11</td><td>0.07</td></tr></table> **BC**	**B1** **B1** **B1** **[3]**	**3.4** **3.4** **1.1**	for 1 correct frequency to 2 d.p. for 3 correct frequencies to 2 d.p. for all 5 correct to 2 d.p. **FT** use of B(4, their 0.19); allow **SC2** for all 5 frequencies awrt 22, 20, 7, 1, 0 or better allow **SC1** for 3 of the above frequencies awrt nearest whole number or better if **B0B0B0** allow **SC1** for probabilities: 0.430467…, 0.403895…, 0.142111…, 0.022223…, 0.001303 given to 3 dp or more; **FT** use of B(4, their 0.19) with $0 < p < 1$
11	(e)	by comparing theoretical and observed frequencies (or probabilities), the model is a (very) poor fit **oe**	**B1** **[1]**	**3.5a**	allow comparison of one pair of frequencies or one pair of probabilities;